

MID SEMESTER**Subject: Machine Learning****Duration: 50 Mins****Answer any four questions out of six**

1. (a) Describe one technique to reduce overfitting. (3)
(b) Explain the role of the decision boundary in logistic regression. (2)
2. (a) Explain the significance of the diagonal line ($AUC = 0.5$) in an ROC curve. What does it indicate about a classifier? (2)
(b) The regularization parameter (C) in SVM controls the trade-off between maximizing the margin and minimizing misclassification. Explain how increasing or decreasing C affects the decision boundary. (3)
3. A decision tree classifier is trained on a dataset containing 30 samples, equally divided between Class A and Class B. After a split, the data is divided into Left Node with 12 samples (8 from Class A, 4 from Class B) and Right Node with 18 samples (7 from Class A, 11 from Class B). Calculate the entropy of the parent node before the split and both child nodes after the split and determine the information gain from the split.
4. A company wants to predict whether a customer will attend a product launch event based on two features: Job Role and Preferred Communication Channel. For a given dataset, if a new customer is a "Manager" and prefers "Email" as their communication channel, use the Naïve Bayes classifier to predict whether they will attend the event.
5. Derive the update rule for Gradient Descent when minimizing a quadratic cost function. Explain how the gradient influences the parameter updates.
6. A medical research team is testing an ML model for diagnosing a disease. The model was tested on 200 patients, where 80 were actually sick. The model predicted 70 patients as sick, out of which 60 were truly sick, while 10 were healthy but misclassified. Additionally, 20 sick patients were misclassified as healthy. Calculate the accuracy, precision, recall, and F1-score of the model.

CID	Job Role	Preferred Communication Channel	Attended Event (Yes/No)
1	Engineer	Phone	No
2	Manager	Email	Yes
3	Developer	Email	No
4	Manager	Email	Yes
5	Engineer	Email	No
6	Manager	Phone	Yes
7	Developer	Phone	No
8	Manager	Email	Yes
9	Engineer	Phone	No
10	Developer	Email	No